

WARAITIP LAOSANGPRATEEP

Engineer · Advanced Materials · Process & Manufacturing · Data-Driven Engineering
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Thailand / Singapore / APAC · Available August 2026



■ PROFILE

Dual-degree engineer combining a Chemical Engineering foundation (Sirindhorn International Institute of Technology (SIIT), Thailand) with a Master's in Medical Engineering (University of Auckland, QS Top 100), underpinned by competitive international research experience across New Zealand and Taiwan. Brings a rare combination of rigorous process engineering fundamentals, hands-on precision materials characterisation, and applied data science, including machine learning model development and statistical validation, that translates directly into high-technology manufacturing, process engineering, and advanced materials roles across Singapore, Thailand, and the broader APAC region. Proven ability to design and execute systematic experimental programmes, derive actionable conclusions from complex datasets, and communicate technical findings with precision. Bilingual Thai–English with strong cross-cultural working experience.

■ EDUCATION

Master of Medical Engineering

Mar 2025 – Jun 2026

University of Auckland · New Zealand · QS World Top 100

- ▶ Thesis: Quantitative mechanical characterisation of neonatal growth plate tissue using nanoindentation and microindentation, generating zone-specific stiffness and hardness profiles across articular cartilage, hypertrophic, proliferative, and resting zones of a structurally heterogeneous biological material
- ▶ Coursework project: Built and validated supervised ML models (SVM classification, cross-validation) in Python to classify meat types from spectral wavelength signatures — applied ML for materials characterisation and process data analysis
- ▶ Coursework: Advanced Functional Materials · Battery · Superconductor · Semiconductor & Materials Science · Medical Device & Technology Development · Advanced Imaging Technologies (OCT, MRI) · Machine Learning for Biomedical Applications · Advanced Drug Disposition & Kinetics · Engineering Project Management

Bachelor of Engineering – Chemical Engineering

Aug 2020 – May 2024

Thammasat University (SIIT) · Thailand · International Programme · GPA 3.22 / 4.00

- ▶ Senior Project: Computational study of butylone- β -cyclodextrin inclusion complex formation using molecular docking (AutoDock) and quantum chemical modelling (Gaussian/GaussView) — presented at PACCON 2024

Core engineering foundation: Chemical Reaction Engineering · Process Design · Plant Design · Thermodynamics · Heat & Mass Transfer · Fluid Mechanics · Environmental Engineering · Wastewater Treatment · Safety in Chemical Operations · Analytical and Instrumental (XRD, FTIR, UV-Vis, Raman, XRF, NMR, AAS, ICP-OES, SEM, TEM, TGA, BET, GC, GC-MS, HPLC, LC-MS)

■ RESEARCH & PROFESSIONAL EXPERIENCE

Master's Thesis Researcher — Materials Characterisation & Data Analysis

Mar 2025 – Present

University of Auckland · Orthopaedic Biomechanics Lab (Dr Ashvin Thambyah) · New Zealand

- ▶ Designed and executed a systematic nanoindentation and microindentation testing programme on neonatal lamb tibia growth plate tissue, producing quantitative elastic modulus and hardness data across four structurally distinct sub-zones
- ▶ Applied statistical testing to identify significant from high-variance datasets; reconciled results against conflicting published literature
- ▶ Independently managed full experimental pipeline: cryogenic tissue preparation, controlled environmental conditions, systematic data acquisition, outlier investigation, and production of publication-quality figures

Research Intern – Nanomaterial Characterisation (SWCNTs)

Jun 2023 – Jul 2023

Academia Sinica — Institute of Atomic and Molecular Sciences (IAMS-IIP) with Prof. Ching-Wei Lin · Taipei, Taiwan [Link](#)

- ▶ Selected for a competitive international internship at one of Asia's leading materials science institutes; characterised Single-Walled Carbon Nanotube (SWCNT) defect populations using absorbance and fluorescence spectroscopy for cancer diagnostic applications
- ▶ Designed and executed a DOE-style experimental matrix; processed multi-dimensional spectral datasets in Origin Lab and contributed structured findings to peer scientific reports
- ▶ Achieved full independent experimental operation within days of arrival, demonstrating rapid technical onboarding and cross-cultural adaptability in an international research environment

Customer Service & Operations (concurrent with full-time postgraduate studies)

Aug 2024 – Jan 2026

Lily's Collection · Blue Elephant · Mi&Chi · Khao San Eatery · The Coffee Club · Auckland, NZ

- ▶ Maintained consistent professional performance across multiple customer-facing and operational roles while simultaneously completing a demanding research-based postgraduate degree, demonstrating punctuality, reliability, multitasking ability, and effective time management through disciplined scheduling and clear task prioritisation that enabled successful delivery on both academic and professional commitments. Adapted quickly to dynamic, fast-paced service environments, applying practical problem-solving and common-sense judgement to resolve on-the-spot operational challenges with minimal supervision.

■ TECHNICAL SKILLS

Process Engineering: Chemical & process fundamentals; thermodynamics; heat & mass transfer; reaction engineering; fluid mechanics; process scale-up; P&ID awareness; Aspen Plus; HYSYS; HAZOP awareness

Materials & Characterisation: Nanoindentation; microindentation; absorbance & fluorescence spectroscopy; nanomaterial characterisation; optical microscopy; mechanical property testing; thin-film material principles

Semiconductor & Mfg: Semiconductor manufacturing fundamentals; wafer fab process awareness; process control principles; yield improvement concepts; SPC; FMEA; CAPA; GMP; factory compliance

Medical Device & Regulatory: Medical device development lifecycle; design verification & validation; ISO 13485; IEC 60601-1; IEC 62366; ISO 14971; FMEA; regulatory documentation; technical report writing

Machine Learning & Data: Python (pandas, NumPy, matplotlib, seaborn, scikit-learn); SVM classification; cross-validation; confusion matrix; supervised ML model development & validation; D3.js; Origin Lab; statistical analysis

Quality & Improvement: DOE (Design of Experiments); SPC; Six Sigma (DMAIC); Lean/Kaizen; FMEA; Fault Tree Analysis; root cause analysis; ISO awareness

Simulation & Computational: Aspen Plus; Gaussian/GaussView; AutoDock; Discovery Studio Visualizer; molecular docking; computational chemistry; drug-material interaction modelling

Biomedical Imaging: OCT; MRI; Ultrasound — operating principles and data interpretation from dedicated postgraduate coursework

■ CERTIFICATIONS & ACHIEVEMENTS

- ▶ Poster Presentation – PACCON 2024 (Pure and Applied Chemistry International Conference)
- ▶ Academia Sinica IAMS-IIP Internship Certificate, Taiwan — international research programme
- ▶ Google Data Analysis with Python Specialisation (Coursera)

■ VOLUNTEER & LEADERSHIP

Narrator & Lab Demonstrator — SIIT Open Days

Jul 2022 – May 2024

Thammasat University (SIIT) · Thailand

- ▶ Designed and facilitated live chemical engineering demonstrations for prospective students; developed clear technical communication skills applicable to customer-site engineering, stakeholder briefing, and cross-functional collaboration

Languages: Thai (Native) · English (IELTS 6.5 · TOEIC 830 — Professional) · Japanese (Elementary)